Supplementary material

All analyses use corrected selection unless labelled historical. Rates and slope units are deaths per 100,000; APC is percent per year. Full-precision CSV results and model warnings are delivered with the code.

Table S1. Corrected selection by year

Year	All SIM	J45/J46	COVID	Duplicates	Invalid dates	Negative age	Valid	≥7	5–34
2014	1227039	2129	0	0	2	0	2127	2035	182
2015	1264175	2232	0	0	1	0	2231	2142	189
2016	1309774	2249	0	0	2	0	2247	2187	184
2017	1312663	2477	0	0	1	0	2476	2418	172
2018	1316719	2307	0	0	4	0	2303	2244	200
2019	1349801	2488	0	0	2	0	2486	2419	218
2020	1556824	2735	27	0	3	0	2705	2677	236
2021	1832649	2488	30	0	0	0	2458	2422	209

COVID = union across five cause lines. Invalid-date and negative-age counts follow COVID and deduplication. Both ambiguous 2018 records are retained. The historical selection, provided separately, removed 56 COVID records, one record under the reduced key and 15 records with invalid dates, leaving 18,583 deaths at continuous age >6.

Table S2. Annual Aligned crude ≥7

Year	Deaths	Population	Rate
2014	2035	181,252,457	1.122743
2015	2142	182,968,415	1.170694
2016	2187	184,647,932	1.184416
2017	2418	186,273,291	1.298093
2018	2244	187,895,171	1.194283
2019	2419	189,505,848	1.276478
2020	2677	191,083,309	1.400960
2021	2422	192,673,672	1.257048

Table S2. Annual Brazil 2014 ASR ≥7

Year	Deaths	Population	Rate
2014	2035	181,252,457	1.122743
2015	2142	182,968,415	1.144431
2016	2187	184,647,932	1.132194
2017	2418	186,273,291	1.208506
2018	2244	187,895,171	1.089633
2019	2419	189,505,848	1.140337
2020	2677	191,083,309	1.229760
2021	2422	192,673,672	1.077177

Table S2. Annual Crude 5–34

Year	Deaths	Population	Rate
2014	182	100,130,699	0.181762
2015	189	99,610,503	0.189739
2016	184	99,048,598	0.185767
2017	172	98,452,312	0.174704
2018	200	97,819,731	0.204458
2019	218	97,214,347	0.224247
2020	236	96,671,200	0.244126
2021	209	96,070,603	0.217548

Table S2. Annual Brazil 2014 ASR 5–34

Year	Deaths	Population	Rate
2014	182	100,130,699	0.181762
2015	189	99,610,503	0.189060
2016	184	99,048,598	0.184945
2017	172	98,452,312	0.174255
2018	200	97,819,731	0.203444
2019	218	97,214,347	0.222528
2020	236	96,671,200	0.241039
2021	209	96,070,603	0.214535

Table S2. Annual Crude 7–34

Year	Deaths	Population	Rate
2014	171	94,235,685	0.181460
2015	174	93,780,394	0.185540
2016	179	93,255,055	0.191947
2017	164	92,641,638	0.177026
2018	190	92,007,559	0.206505
2019	208	91,371,402	0.227642
2020	232	90,729,128	0.255706
2021	201	90,129,917	0.223011

Table S2. Annual Crude 35–59

Year	Deaths	Population	Rate
2014	452	62,893,940	0.718670
2015	485	64,149,669	0.756044
2016	515	65,398,428	0.787481
2017	567	66,640,781	0.850830

Year	Deaths	Population	Rate
2018	504	67,862,310	0.742680
2019	524	69,039,371	0.758987
2020	709	70,157,104	1.010589
2021	591	71,213,520	0.829899

Table S2. Annual Crude ≥60

Year	Deaths	Population	Rate
2014	1412	24,122,832	5.853376
2015	1483	25,038,352	5.922914
2016	1493	25,994,449	5.743534
2017	1687	26,990,872	6.250261
2018	1550	28,025,302	5.530716
2019	1687	29,095,075	5.798232
2020	1736	30,197,077	5.748901
2021	1630	31,330,235	5.202642

Table S3. Historical age-group reproduction

series	n_2014	n_2021	rate_2014	rate_2021	pct_change_2014_2021	slope	p_value
legacy_lt18	43	37	0.07649	0.06945	-9.19335	-0.00047	0.85236
legacy_18_59	580	756	0.47786	0.58735	22.91210	0.02125	0.04388
legacy_gte60	1412	1630	5.85338	5.20264	-11.11724	-0.07124	0.13500

Historical <18 numerator is floor ages 7–17 and denominator ages 0–17. These rows reproduce the historical construction and are not corrected primary domains.

Table S4. Count models, IBGE 2018

Series	Family	APC	95% CI	p	Dispersion	Theta	Status
Crude ≥7	Poisson	2.190	1.549, 2.835	<0.001	6.04		ok
Crude ≥7	Quasi-Poisson	2.190	0.236, 4.183	0.033	6.04		ok
Crude ≥7	Negative binomial	2.221	0.886, 3.574	0.001	1.33	686.8	ok
Crude 5–34	Poisson	3.998	1.787, 6.258	<0.001	1.09		ok
Crude 5–34	Quasi-Poisson	3.998	1.130, 6.949	0.014	1.09		ok
Crude 5–34	Negative binomial	3.998	1.787, 6.258	<0.001	1.09	3280138.3	warning
Crude 7–34	Poisson	4.604	2.327, 6.931	<0.001	1.15		ok
Crude 7–34	Quasi-Poisson	4.604	1.561, 7.738	0.010	1.15		ok
Crude 7–34	Negative binomial	4.604	2.327, 6.931	<0.001	1.15	2427388.1	warning
Crude 35–59	Poisson	2.779	1.448, 4.126	<0.001	5.54		ok
Crude 35–59	Quasi-Poisson	2.779	-1.082, 6.789	0.130	5.54		ok
Crude 35–59	Negative binomial	2.783	0.181, 5.453	0.036	1.34	187.9	ok

Series	Family	APC	95% CI	p	Dispersion	Theta	Status
Crude ≥60	Poisson	-1.300	-2.047, -0.546	<0.001	3.55		ok
Crude ≥60	Quasi-Poisson	-1.300	-3.049, 0.482	0.124	3.55		ok
Crude ≥60	Negative binomial	-1.271	-2.480, -0.046	0.042	1.34	972.6	ok
Age-adjusted ≥7	Poisson	0.001	-0.626, 0.632	0.998	3.11		ok
Age-adjusted ≥7	Quasi-Poisson	0.001	-1.125, 1.140	0.999	3.11		ok
Age-adjusted ≥7	Negative binomial	0.926	-0.197, 2.061	0.106	1.30	147.8	ok

Rows marked as warnings reached the theta iteration limit; convergence of mean-model IRLS does not prove convergence of theta. Retain these estimates with caution. The recorded warning was “iteration limit reached”. All errors, warnings, iterations, coefficients and fitted objects are retained in CSV/RDS. Quasi-Poisson uses t intervals with residual degrees of freedom; Poisson and negative binomial use normal Wald intervals.

Table S4. Count models, IBGE 2024

Series	Family	APC	95% CI	p	Dispersion	Theta	Status
Crude ≥7	Poisson	2.312	1.670, 2.958	<0.001	5.98		ok
Crude ≥7	Quasi-Poisson	2.312	0.366, 4.297	0.027	5.98		ok
Crude ≥7	Negative binomial	2.342	1.012, 3.690	<0.001	1.33	695.6	ok
Crude 5–34	Poisson	4.178	1.964, 6.441	<0.001	1.09		ok
Crude 5–34	Quasi-Poisson	4.178	1.303, 7.135	0.012	1.09		ok
Crude 5–34	Negative binomial	4.178	1.964, 6.441	<0.001	1.09	3258238.9	warning
Crude 7–34	Poisson	4.791	2.511, 7.122	<0.001	1.16		ok
Crude 7–34	Quasi-Poisson	4.791	1.742, 7.931	0.008	1.16		ok
Crude 7–34	Negative binomial	4.791	2.511, 7.122	<0.001	1.16	2402052.5	warning
Crude 35–59	Poisson	2.843	1.511, 4.192	<0.001	5.52		ok
Crude 35–59	Quasi-Poisson	2.843	-1.013, 6.849	0.123	5.52		ok
Crude 35–59	Negative binomial	2.844	0.246, 5.510	0.032	1.34	189.0	ok
Crude ≥60	Poisson	-1.210	-1.959, -0.455	0.002	3.33		ok
Crude ≥60	Quasi-Poisson	-1.210	-2.907, 0.517	0.136	3.33		ok
Crude ≥60	Negative binomial	-1.186	-2.359, 0.001	0.050	1.34	1080.7	ok
Age-adjusted ≥7	Poisson	0.290	-0.340, 0.924	0.368	3.02		ok
Age-adjusted ≥7	Quasi-Poisson	0.290	-0.823, 1.415	0.606	3.02		ok
Age-adjusted ≥7	Negative binomial	1.137	0.031, 2.256	0.044	1.31	156.0	ok

Rows marked as warnings reached the theta iteration limit; convergence of mean-model IRLS does not prove convergence of theta. Retain these estimates with caution. The recorded warning was “iteration limit reached”. All errors, warnings, iterations, coefficients and fitted objects are retained in CSV/RDS. Quasi-Poisson uses t intervals with residual degrees of freedom; Poisson and negative binomial use normal Wald intervals.

Table S5. Standards and domains, IBGE 2018

Series	2014	2021	Δ rate	Δ %	Slope (95% CI)	p
Brazil ASR 35–59	0.71867	0.82485	+0.10618	+14.77	0.02134 (-0.00957, 0.05225)	0.142
Brazil ASR 5–34	0.18176	0.21453	+0.03277	+18.03	0.00751 (0.00181, 0.01322)	0.018

Series	2014	2021	Δ rate	Δ %	Slope (95% CI)	p
Brazil ASR 7–34	0.18146	0.21955	+0.03809	+20.99	0.00875 (0.00260, 0.01490)	0.013
Brazil ASR ≥60	5.85338	5.08539	-0.76799	-13.12	-0.08864 (-0.18714, 0.00986)	0.070
Brazil ASR ≥7	1.12274	1.07718	-0.04557	-4.06	0.00016 (-0.02139, 0.02170)	0.986
WHO ASR 35–59	0.71921	0.82668	+0.10747	+14.94	0.02162 (-0.00936, 0.05260)	0.139
WHO ASR 5–34	0.17788	0.20702	+0.02914	+16.38	0.00678 (0.00140, 0.01216)	0.022
WHO ASR 7–34	0.17730	0.21246	+0.03516	+19.83	0.00819 (0.00235, 0.01403)	0.014
WHO ASR ≥60	5.55943	4.81550	-0.74393	-13.38	-0.08637 (-0.17725, 0.00451)	0.059
WHO ASR ≥7	1.09451	1.04795	-0.04656	-4.25	-0.00015 (-0.02077, 0.02047)	0.986

Table S5. Standards and domains, IBGE 2024

Series	2014	2021	Δ rate	Δ %	Slope (95% CI)	p
Brazil ASR 35–59	0.72112	0.83140	+0.11028	+15.29	0.02192 (-0.00907, 0.05291)	0.134
Brazil ASR 5–34	0.18281	0.21972	+0.03691	+20.19	0.00819 (0.00232, 0.01406)	0.014
Brazil ASR 7–34	0.18258	0.22505	+0.04247	+23.26	0.00947 (0.00313, 0.01581)	0.011
Brazil ASR ≥60	5.88365	5.30051	-0.58314	-9.91	-0.06471 (-0.16221, 0.03278)	0.155
Brazil ASR ≥7	1.12844	1.11123	-0.01722	-1.53	0.00393 (-0.01762, 0.02547)	0.671
WHO ASR 35–59	0.72095	0.83282	+0.11187	+15.52	0.02223 (-0.00879, 0.05326)	0.130
WHO ASR 5–34	0.17884	0.21201	+0.03317	+18.55	0.00744 (0.00193, 0.01294)	0.016
WHO ASR 7–34	0.17835	0.21778	+0.03943	+22.11	0.00889 (0.00290, 0.01487)	0.011
WHO ASR ≥60	5.60811	5.01048	-0.59763	-10.66	-0.06712 (-0.15761, 0.02337)	0.119
WHO ASR ≥7	1.10228	1.07940	-0.02288	-2.08	0.00305 (-0.01761, 0.02370)	0.731

Table S6. Kitagawa, IBGE 2018

Endpoints	Crude start	Crude end	Net Δ	Composition	Rate term	Composition %
2014–2021	1.122743	1.257048	+0.134304	+0.191434	-0.057129	142.54
2014–2019	1.122743	1.276478	+0.153734	+0.138122	+0.015612	89.84
2014–2020	1.122743	1.400960	+0.278216	+0.171157	+0.107059	61.52
2015–2021	1.170694	1.257048	+0.086354	+0.167391	-0.081037	193.84
2016–2021	1.184416	1.257048	+0.072632	+0.139935	-0.067304	192.66
2014–2018	1.122743	1.194283	+0.071540	+0.107366	-0.035827	150.08

Each term is independently rounded. The unrounded identity is tested to 1e−12; rounded terms may differ in their last displayed digit. A share over 100% reflects a negative rate term, not a probability.

Table S6. Kitagawa, IBGE 2024

Endpoints	Crude start	Crude end	Net Δ	Composition	Rate term	Composition %
2014–2021	1.128443	1.274118	+0.145675	+0.170443	-0.024768	117.00
2014–2019	1.128443	1.290082	+0.161639	+0.130868	+0.030771	80.96
2014–2020	1.128443	1.417122	+0.288679	+0.158796	+0.129883	55.01
2015–2021	1.177572	1.274118	+0.096546	+0.148180	-0.051634	153.48
2016–2021	1.192574	1.274118	+0.081544	+0.121796	-0.040252	149.36
2014–2018	1.128443	1.205894	+0.077451	+0.101345	-0.023893	130.85

Each term is independently rounded. The unrounded identity is tested to 1e−12; rounded terms may differ in their last displayed digit. A share over 100% reflects a negative rate term, not a probability.

Table S7. Pandemic-period influence, age_5_34, IBGE 2018

Scenario	Series	Years	Δ %	Slope (95% CI)	p
Principal	Brazil ASR 5–34	8	+18.03	0.00751 (0.00181, 0.01322)	0.018
Principal	Crude 5–34	8	+19.69	0.00795 (0.00212, 0.01377)	0.016
Principal	WHO ASR 5–34	8	+16.38	0.00678 (0.00140, 0.01216)	0.022
Without 2020	Brazil ASR 5–34	7	+18.03	0.00581 (0.00040, 0.01122)	0.040
Without 2020	Crude 5–34	7	+19.69	0.00619 (0.00070, 0.01167)	0.034
Without 2020	WHO ASR 5–34	7	+16.38	0.00536 (-0.00017, 0.01089)	0.055
2014–2019	Brazil ASR 5–34	6	+22.43	0.00675 (-0.00228, 0.01578)	0.107
2014–2019	Crude 5–34	6	+23.37	0.00701 (-0.00219, 0.01622)	0.102
2014–2019	WHO ASR 5–34	6	+22.80	0.00660 (-0.00250, 0.01570)	0.114

Without 2020 uses 2014–2019 and 2021 as calendar years (seven observations, 5 residual df); prepandemic uses 2014–2019 (six observations, 4 residual df). Endpoint contrasts are distinct from annual slopes. Count-model sensitivities for the same domains are supplied below.

Table S7 continued. Count models, age_5_34, IBGE 2018

Scenario	Family	APC	95% CI	p	df	Status
Principal	Poisson	3.998	1.787, 6.258	<0.001	6	ok
Principal	Quasi-Poisson	3.998	1.130, 6.949	0.014	6	ok
Principal	Negative binomial	3.998	1.787, 6.258	<0.001	6	warning
Without 2020	Poisson	3.159	0.743, 5.633	0.010	5	ok
Without 2020	Quasi-Poisson	3.159	0.382, 6.012	0.033	5	ok
Without 2020	Negative binomial	3.159	0.743, 5.633	0.010	5	warning
2014–2019	Poisson	3.673	0.212, 7.253	0.037	4	ok
2014–2019	Quasi-Poisson	3.673	-0.992, 8.558	0.095	4	ok
2014–2019	Negative binomial	3.673	0.212, 7.253	0.037	4	warning

Table S7. Pandemic-period influence, age_7_34, IBGE 2018

Scenario	Series	Years	Δ %	Slope (95% CI)	p
Principal	Brazil ASR 7–34	8	+20.99	0.00875 (0.00260, 0.01490)	0.013
Principal	Crude 7–34	8	+22.90	0.00926 (0.00294, 0.01559)	0.012
Principal	WHO ASR 7–34	8	+19.83	0.00819 (0.00235, 0.01403)	0.014
Without 2020	Brazil ASR 7–34	7	+20.99	0.00666 (0.00152, 0.01180)	0.021
Without 2020	Crude 7–34	7	+22.90	0.00710 (0.00188, 0.01232)	0.017
Without 2020	WHO ASR 7–34	7	+19.83	0.00633 (0.00108, 0.01158)	0.027
2014–2019	Brazil ASR 7–34	6	+24.38	0.00767 (-0.00086, 0.01620)	0.067
2014–2019	Crude 7–34	6	+25.45	0.00797 (-0.00076, 0.01670)	0.064
2014–2019	WHO ASR 7–34	6	+25.20	0.00768 (-0.00087, 0.01622)	0.067

Without 2020 uses 2014–2019 and 2021 as calendar years (seven observations, 5 residual df); prepandemic uses 2014–2019 (six observations, 4 residual df). Endpoint contrasts are distinct from annual slopes. Count-model sensitivities for the same domains are supplied below.

Table S7 continued. Count models, age_7_34, IBGE 2018

Scenario	Family	APC	95% CI	p	df	Status
Principal	Poisson	4.604	2.327, 6.931	<0.001	6	ok
Principal	Quasi-Poisson	4.604	1.561, 7.738	0.010	6	ok
Principal	Negative binomial	4.604	2.327, 6.931	<0.001	6	warning
Without 2020	Poisson	3.590	1.104, 6.138	0.004	5	ok
Without 2020	Quasi-Poisson	3.590	0.975, 6.274	0.016	5	ok
Without 2020	Negative binomial	3.590	1.104, 6.138	0.004	5	warning
2014–2019	Poisson	4.153	0.584, 7.849	0.022	4	ok
2014–2019	Quasi-Poisson	4.153	-0.220, 8.718	0.058	4	ok
2014–2019	Negative binomial	4.153	0.583, 7.849	0.022	4	warning

Table S7. Pandemic-period influence, age_5_34, IBGE 2024

Scenario	Series	Years	Δ %	Slope (95% CI)	p
Principal	Brazil ASR 5–34	8	+20.19	0.00819 (0.00232, 0.01406)	0.014
Principal	Crude 5–34	8	+21.07	0.00841 (0.00249, 0.01433)	0.013
Principal	WHO ASR 5–34	8	+18.55	0.00744 (0.00193, 0.01294)	0.016
Without 2020	Brazil ASR 5–34	7	+20.19	0.00643 (0.00087, 0.01199)	0.031
Without 2020	Crude 5–34	7	+21.07	0.00662 (0.00105, 0.01220)	0.028
Without 2020	WHO ASR 5–34	7	+18.55	0.00596 (0.00036, 0.01155)	0.041
2014–2019	Brazil ASR 5–34	6	+24.27	0.00740 (-0.00188, 0.01669)	0.091
2014–2019	Crude 5–34	6	+24.69	0.00754 (-0.00180, 0.01687)	0.088
2014–2019	WHO ASR 5–34	6	+24.37	0.00720 (-0.00201, 0.01642)	0.096

Without 2020 uses 2014–2019 and 2021 as calendar years (seven observations, 5 residual df); prepandemic uses 2014–2019 (six observations, 4 residual df). Endpoint contrasts are distinct from annual slopes. Count-model sensitivities for the same domains are supplied below.

Table S7 continued. Count models, age_5_34, IBGE 2024

Scenario	Family	APC	95% CI	p	df	Status
Principal	Poisson	4.178	1.964, 6.441	<0.001	6	ok
Principal	Quasi-Poisson	4.178	1.303, 7.135	0.012	6	ok
Principal	Negative binomial	4.178	1.964, 6.441	<0.001	6	warning
Without 2020	Poisson	3.339	0.919, 5.816	0.007	5	ok
Without 2020	Quasi-Poisson	3.339	0.551, 6.203	0.027	5	ok
Without 2020	Negative binomial	3.339	0.919, 5.816	0.007	5	warning
2014–2019	Poisson	3.899	0.431, 7.487	0.027	4	ok
2014–2019	Quasi-Poisson	3.899	-0.768, 8.786	0.082	4	ok
2014–2019	Negative binomial	3.899	0.431, 7.487	0.027	4	warning

Table S7. Pandemic-period influence, age_7_34, IBGE 2024

Scenario	Series	Years	Δ %	Slope (95% CI)	p
Principal	Brazil ASR 7–34	8	+23.26	0.00947 (0.00313, 0.01581)	0.011
Principal	Crude 7–34	8	+24.37	0.00977 (0.00334, 0.01620)	0.010
Principal	WHO ASR 7–34	8	+22.11	0.00889 (0.00290, 0.01487)	0.011
Without 2020	Brazil ASR 7–34	7	+23.26	0.00732 (0.00202, 0.01262)	0.016
Without 2020	Crude 7–34	7	+24.37	0.00757 (0.00225, 0.01288)	0.015
Without 2020	WHO ASR 7–34	7	+22.11	0.00696 (0.00164, 0.01227)	0.020
2014–2019	Brazil ASR 7–34	6	+26.32	0.00837 (-0.00043, 0.01716)	0.057
2014–2019	Crude 7–34	6	+26.85	0.00853 (-0.00033, 0.01739)	0.056
2014–2019	WHO ASR 7–34	6	+26.85	0.00831 (-0.00035, 0.01697)	0.056

Without 2020 uses 2014–2019 and 2021 as calendar years (seven observations, 5 residual df); prepandemic uses 2014–2019 (six observations, 4 residual df). Endpoint contrasts are distinct from annual slopes. Count-model sensitivities for the same domains are supplied below.

Table S7 continued. Count models, age_7_34, IBGE 2024

Scenario	Family	APC	95% CI	p	df	Status
Principal	Poisson	4.791	2.511, 7.122	<0.001	6	ok
Principal	Quasi-Poisson	4.791	1.742, 7.931	0.008	6	ok
Principal	Negative binomial	4.791	2.511, 7.122	<0.001	6	warning
Without 2020	Poisson	3.778	1.288, 6.329	0.003	5	ok
Without 2020	Quasi-Poisson	3.778	1.150, 6.474	0.014	5	ok
Without 2020	Negative binomial	3.778	1.288, 6.329	0.003	5	warning
2014–2019	Poisson	4.391	0.814, 8.096	0.016	4	ok
2014–2019	Quasi-Poisson	4.391	0.018, 8.956	0.049	4	ok
2014–2019	Negative binomial	4.391	0.814, 8.096	0.016	4	warning

Table S8. One-condition sensitivities, ge7, IBGE 2018

Scenario	Deaths	2014	2021	Δ %	Slope (95% CI)	p
Principal	18544	1.12274	1.07718	-4.06	0.00016 (-0.02139, 0.02170)	0.986
Days/365.25	18544	1.12274	1.07719	-4.06	0.00016 (-0.02139, 0.02170)	0.986
Original COVID	18545	1.12274	1.07772	-4.01	0.00020 (-0.02130, 0.02171)	0.982
Retain COVID	18601	1.12274	1.09149	-2.78	0.00210 (-0.01965, 0.02385)	0.821
Historical key	18543	1.12274	1.07718	-4.06	0.00015 (-0.02142, 0.02172)	0.987

Each scenario changes only the indicated condition from corrected selection. Calendar-boundary changes are recorded individually in record_selection_and_age_transitions.csv; COVID overlaps in covid_overlap_patterns.csv.

Table S8. One-condition sensitivities, age_5_34, IBGE 2018

Scenario	Deaths	2014	2021	Δ %	Slope (95% CI)	p
Principal	1590	0.18176	0.21453	+18.03	0.00751 (0.00181, 0.01322)	0.018

Scenario	Deaths	2014	2021	Δ %	Slope (95% CI)	p
Days/365.25	1590	0.18176	0.21453	+18.03	0.00751 (0.00181, 0.01322)	0.018
Original COVID	1591	0.18176	0.21553	+18.58	0.00760 (0.00195, 0.01324)	0.017
Retain COVID	1596	0.18176	0.21960	+20.82	0.00800 (0.00246, 0.01353)	0.012
Historical key	1590	0.18176	0.21453	+18.03	0.00751 (0.00181, 0.01322)	0.018

Each scenario changes only the indicated condition from corrected selection. Calendar-boundary changes are recorded individually in record_selection_and_age_transitions.csv; COVID overlaps in covid_overlap_patterns.csv.

Table S8. One-condition sensitivities, age_7_34, IBGE 2018

Scenario	Deaths	2014	2021	Δ %	Slope (95% CI)	p
Principal	1519	0.18146	0.21955	+20.99	0.00875 (0.00260, 0.01490)	0.013
Days/365.25	1519	0.18146	0.21955	+20.99	0.00875 (0.00260, 0.01490)	0.013
Original COVID	1520	0.18146	0.22060	+21.57	0.00884 (0.00275, 0.01493)	0.012
Retain COVID	1525	0.18146	0.22493	+23.96	0.00926 (0.00331, 0.01522)	0.009
Historical key	1519	0.18146	0.21955	+20.99	0.00875 (0.00260, 0.01490)	0.013

Each scenario changes only the indicated condition from corrected selection. Calendar-boundary changes are recorded individually in record_selection_and_age_transitions.csv; COVID overlaps in covid_overlap_patterns.csv.

Table S8. One-condition sensitivities, ge7, IBGE 2024

Scenario	Deaths	2014	2021	Δ %	Slope (95% CI)	p
Principal	18544	1.12844	1.11123	-1.53	0.00393 (-0.01762, 0.02547)	0.671
Days/365.25	18544	1.12844	1.11123	-1.53	0.00393 (-0.01762, 0.02547)	0.671
Original COVID	18545	1.12844	1.11178	-1.48	0.00397 (-0.01753, 0.02547)	0.667
Retain COVID	18601	1.12844	1.12583	-0.23	0.00590 (-0.01590, 0.02769)	0.532
Historical key	18543	1.12844	1.11123	-1.53	0.00392 (-0.01765, 0.02549)	0.672

Each scenario changes only the indicated condition from corrected selection. Calendar-boundary changes are recorded individually in record_selection_and_age_transitions.csv; COVID overlaps in covid_overlap_patterns.csv.

Table S8. One-condition sensitivities, age_5_34, IBGE 2024

Scenario	Deaths	2014	2021	Δ %	Slope (95% CI)	p
Principal	1590	0.18281	0.21972	+20.19	0.00819 (0.00232, 0.01406)	0.014
Days/365.25	1590	0.18281	0.21972	+20.19	0.00819 (0.00232, 0.01406)	0.014
Original COVID	1591	0.18281	0.22073	+20.74	0.00827 (0.00246, 0.01409)	0.013
Retain COVID	1596	0.18281	0.22498	+23.06	0.00869 (0.00299, 0.01439)	0.010
Historical key	1590	0.18281	0.21972	+20.19	0.00819 (0.00232, 0.01406)	0.014

Each scenario changes only the indicated condition from corrected selection. Calendar-boundary changes are recorded individually in record_selection_and_age_transitions.csv; COVID overlaps in covid_overlap_patterns.csv.

Table S8. One-condition sensitivities, age_7_34, IBGE 2024

Scenario	Deaths	2014	2021	Δ %	Slope (95% CI)	p
Principal	1519	0.18258	0.22505	+23.26	0.00947 (0.00313, 0.01581)	0.011
Days/365.25	1519	0.18258	0.22505	+23.26	0.00947 (0.00313, 0.01581)	0.011

Scenario	Deaths	2014	2021	Δ %	Slope (95% CI)	p
Original COVID	1520	0.18258	0.22612	+23.85	0.00956 (0.00329, 0.01583)	0.010
Retain COVID	1525	0.18258	0.23063	+26.32	0.01000 (0.00387, 0.01614)	0.007
Historical key	1519	0.18258	0.22505	+23.26	0.00947 (0.00313, 0.01581)	0.011

Each scenario changes only the indicated condition from corrected selection. Calendar-boundary changes are recorded individually in record_selection_and_age_transitions.csv; COVID overlaps in covid_overlap_patterns.csv.

Supplementary Figure S1

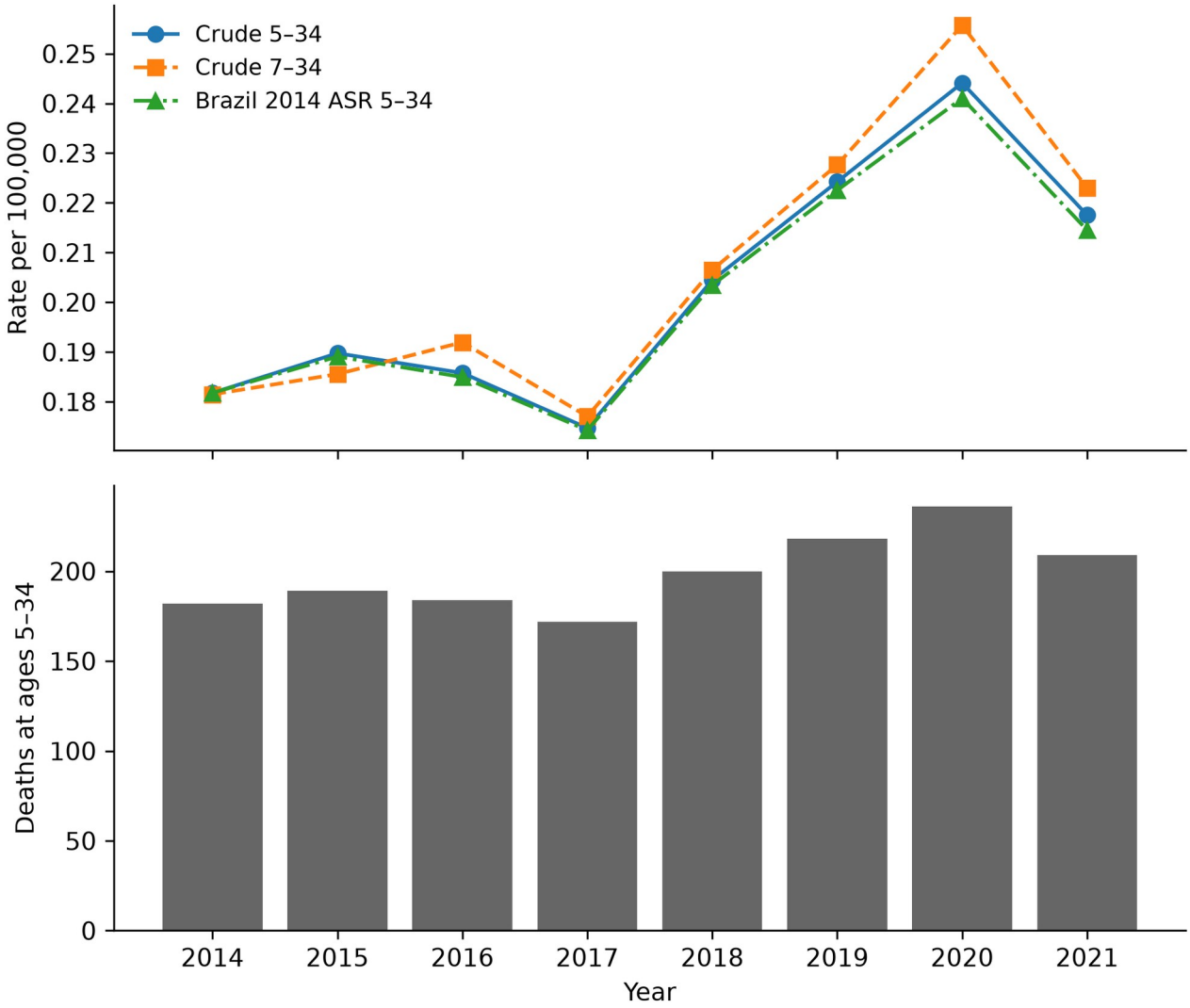


Figure S1. Annual crude rates for ages 5–34 and 7–34, within-5–34 Brazil-2014-standardised rates, and 5–34 death counts; corrected selection and IBGE 2018.

Supplementary methods and provenance

The historical deduplication key combined parsed death date, elapsed-days age, underlying cause and the five cause lines. It collapsed 2018 source rows 971828 and 971831. Inspection of all public fields found differences in occupation and seven administrative fields (NUMEROLOTE, TPPOS, DTCADASTRO, DTRECEBIM, DTRECORIGA, DIFDATA and CONTADOR). The SIM dictionary identifies NUMERODO as the death-declaration number and describes registration and receipt dates as administrative information [6]. NUMERODO is absent from these CSVs. The available fields therefore do not establish whether the rows refer to the same death. We retained both in the primary analysis and assessed their collapse under the historical key as a sensitivity.

Completed age was the difference between death and birth years, reduced by one if death preceded that year's birthday. For births on 29 February, the anniversary in a non-leap year was 1 March. We added a leading zero to seven-digit day-month-year strings and required an eight-digit valid calendar date. Missing or uninterpretable dates and birth after death were excluded and logged separately. We applied no upper-age exclusion and grouped ages ≥90 into the IBGE 90+ interval.

We reconciled the archived 12-page manuscript of 10 October 2026 with a Word copy of concordant content and repository commit 3884c868a07164710a2e358180b3034f9cdcf642. Official SIM archives recovered on 11 October 2026 reproduced the versioned selection counts and year-age aggregates exactly. Historical execution hashes were available for SIM 2020–2021 only, so byte identity with the historical 2014–2019 files could not be established. The accompanying provenance manifest records the reference PDF hash and all verified input hashes.

National single-year populations for both sexes came from the Brazilian Institute of Geography and Statistics (IBGE) 2018 revision [7]; the 2024 revision following the 2022 Census provided a sensitivity [8]. We used the total-population block on sheet BR in the 2018 workbook and SIGLA = BR with SEXO = Ambos in the 2024 workbook. Both provide ages 0–89 and an open 90+ interval. Corrected crude rates matched each numerator to the same population age range. To reproduce the historical construction, we divided continuous-age >6 deaths by the entire population and separately by population ≥7. The latter comparator retains the small mismatch between its historical numerator and denominator.

We used R 4.6.1 [12] for reproduction and revision; the reference reported R 4.3.3. The code records package versions, configurations, input hashes, warnings and independent checks, including numerical differences under the current environment. Public inputs, verified extraction code and separate scenario outputs support reproduction. The repository address is supplied separately to the editors.